

1. Introduction & Motivation

Next POI Recommendation: predict the next place a user will visit, given historical check-in trajectories from Location-Based Social Networks (LBSN).

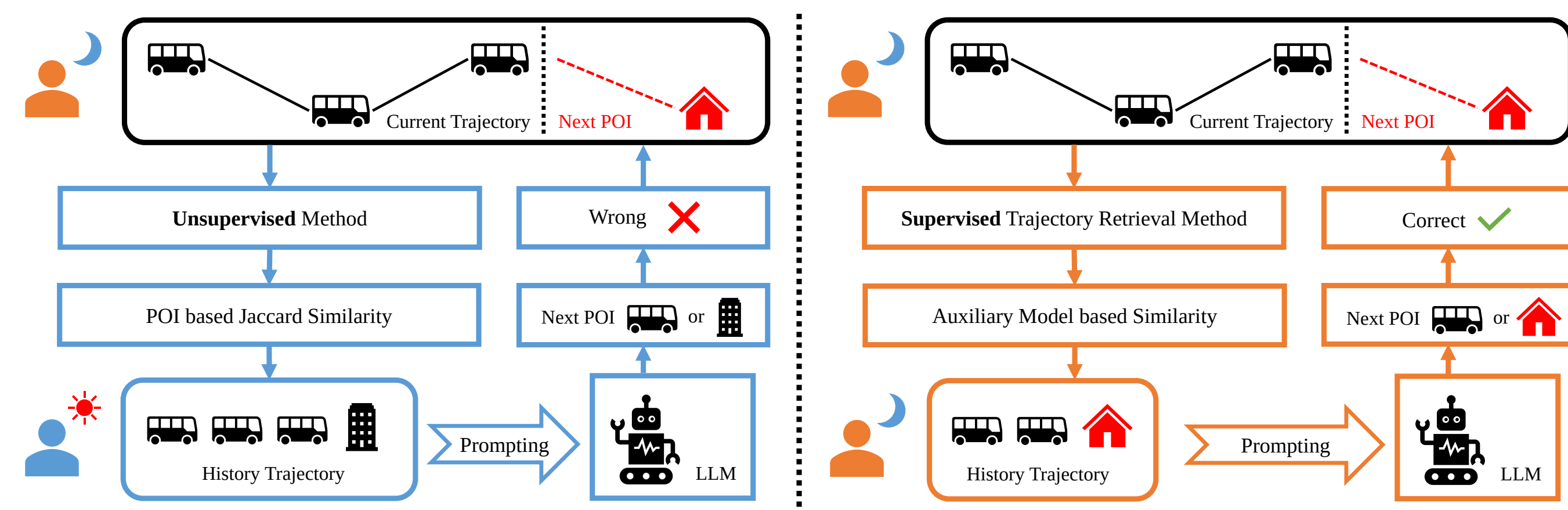
LLM-based paradigm: cast the task as question answering. Existing methods (e.g., *LLM4POI*) still suffer from:

- ✗ **Unsupervised retrieval** may fail to put the correct next POI into the prompt, hindering in-context learning.
- ✗ **Insufficient travel semantics** + sample-based generation yields duplicate outputs.

2. Key Contributions

- **Supervised auxiliary task** fusing heterogeneous similarities (POI / user / category / temporal).
- **Travel semantic-informed candidate module** injecting nearest & most popular POIs.
- **Beam-search generation** for diverse outputs, enabling Acc@1/5/10 evaluation.
- **SOTA on three public datasets**, 3%–32% improvement.

3. Core Idea

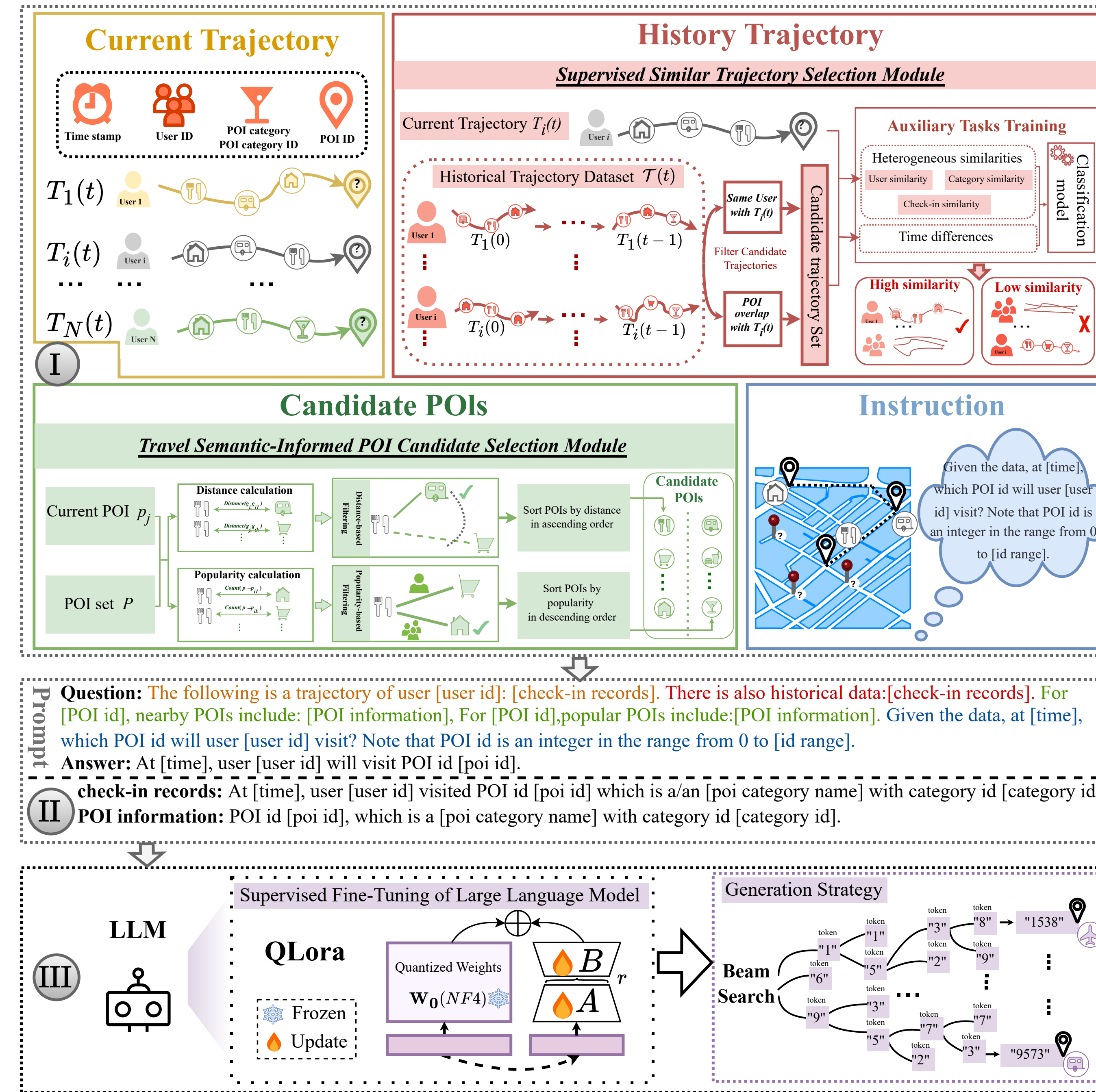


Unsupervised similarity vs. supervised retrieval.

4. Datasets

Dataset	#User	#POI	#Cat.	#Check-in	Trajectory (train/val/test)
Foursquare-NYC	1048	4981	318	103,941	11022/1400/1347
Foursquare-TKY	2282	7833	290	405,000	51661/6868/7038
Gowalla-CA	3957	9690	296	238,369	36374/3529/2780

5. Framework Overview



Pipeline: (I) augmented prompting → (II) SFT of LLM → (III) beam-search generation.

6. Methodology

Augmented Prompting. Current Trajectory | Historical Trajectory | Candidate POIs | Instruction.

Supervised Trajectory Selection. An auxiliary classifier f_w predicts whether T_j contains T_i 's next POI:

$$\min_w \sum_{i,j} l(f_w(T_i, T_j), y_{ij})$$

Features: check-in, user, category, hour, weekday similarities + time difference.

Travel Semantic Candidates.

Nearest POIs: $D(T_i) = \arg \min_j \text{Dist}(g_{ik}, g_j)$.

Popular POIs: $\text{Pop}(T_i) = \arg \max_j \text{Count}(p_{ik} \rightarrow p_j)$.

SFT + Beam Search. LoRA + NF4 quantization.

7. Main Results

	Foursquare-NYC			Foursquare-TKY			Gowalla-CA		
	Acc@1	Acc@5	Acc@10	Acc@1	Acc@5	Acc@10	Acc@1	Acc@5	Acc@10
STGCN	0.1799	0.3425	0.4279	0.1716	0.3453	0.3927	0.0961	0.2097	0.2163
PLSPL	0.1917	0.3678	0.4523	0.1889	0.3523	0.4150	0.1072	0.2278	0.2995
STAN	0.2231	0.4582	0.5734	0.1963	0.3798	0.4464	0.1104	0.2348	0.3018
GETNext	0.2435	0.5089	0.6143	0.2254	0.4417	0.5287	0.1357	0.2852	0.3590
STHGCN	0.2734	0.5361	0.6244	0.2950	0.5207	0.5980	0.1730	0.3529	0.4191
LLM4POI	0.3372	N/A	N/A	0.3035	N/A	N/A	0.2065	N/A	N/A
LLMMob	0.2764	0.5515	0.6330	0.2218	0.4789	0.5529	0.1772	0.3553	0.4217
STrajRAG w/o BS	0.3552	N/A	N/A	0.3195	N/A	N/A	0.2346	N/A	N/A
STrajRAG-GBM	<u>0.3563</u>	<u>0.5949</u>	<u>0.6560</u>	<u>0.3295</u>	<u>0.5364</u>	<u>0.6006</u>	<u>0.2554</u>	<u>0.4528</u>	<u>0.5238</u>
STrajRAG	0.3592	0.6044	0.6719	0.3389	0.5470	0.6135	0.2596	0.4684	0.5429
Improve(%)	6.5	9.6	6.2	11.7	5.1	2.6	25.7	31.8	28.7

8. Ablation (NYC, Acc@1 / Acc@10)

Variant	Acc@1	Acc@10
Full model	0.3592	0.6719
w/o user similarity	0.3364	0.6300
w/o check-in similarity	0.3519	0.6684
w/o distance candidate	0.3595	0.6692
w/o popularity candidate	0.3595	0.6698

9. Efficiency (NYC)

Model	Train	Inference
STHGCN	36 min	0 min
STrajRAG-Qwen2.5-0.5B	23 min	5 min
STrajRAG-Qwen2.5-0.5B-distill	143 min	1 min
STrajRAG-Llama2-7B	371 min	13 min

10. Conclusion

STrajRAG enhances LLM-based next POI recommendation via **supervised trajectory retrieval**, **travel semantic candidate selection**, and **beam search**. It consistently outperforms SOTA baselines on three datasets and enables Acc@5/Acc@10 evaluation. Future work: efficiency (distillation, token optimization), and transfer to trajectory completion and trajectory-user linking.